

Connector from gate drivers

Includes multiple redundant SSO (six-switch-off) pathways:

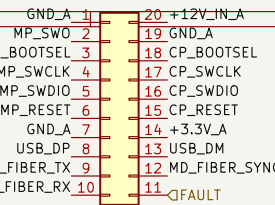
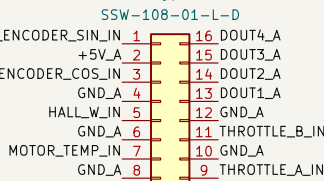
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2

(Gate driver power enable implements 1002, with monitoring feedback).

feedback pins are 0-3.3v binary (3.3v expected when +12v present)

Connector to IO board

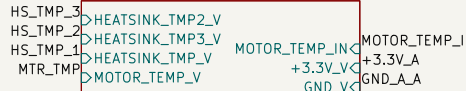
Digital Outputs provided here to IOBoard.



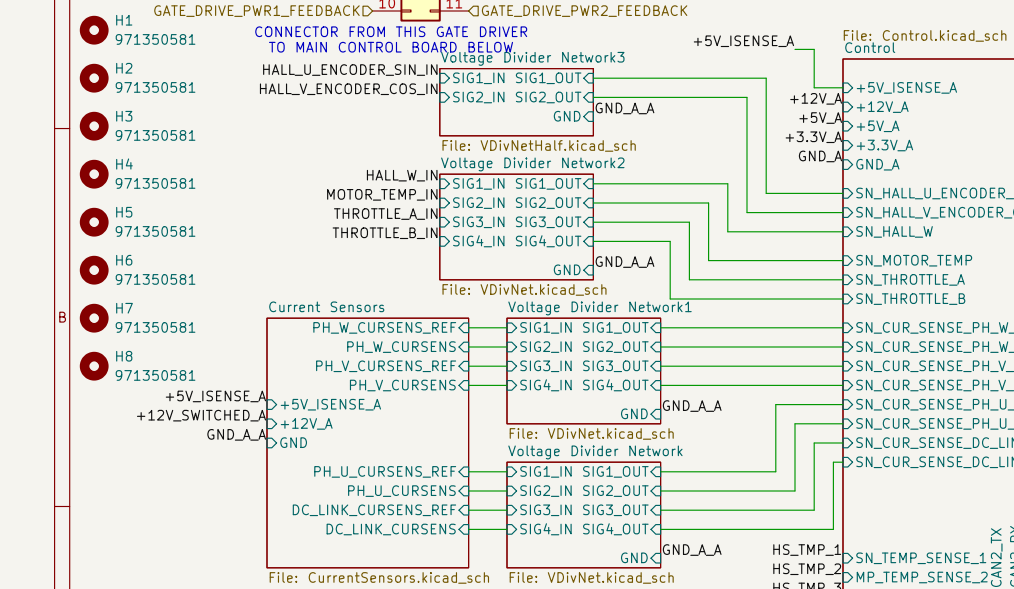
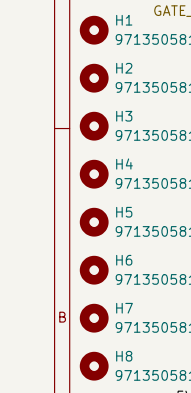
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)

Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.

Temperature Sensors



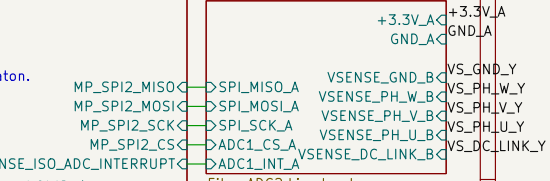
File: TemperatureSensors.kicad_sch



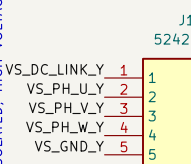
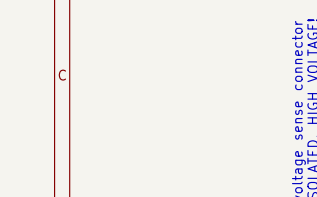
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults

Please see HARA under documentation for additional informaton.

ADC

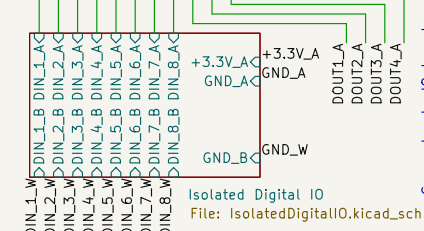
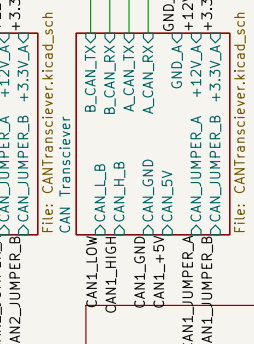
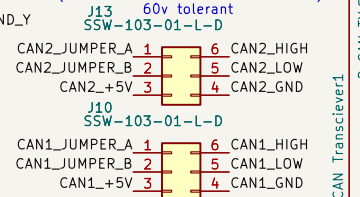


File: ADC2.kicad_sch

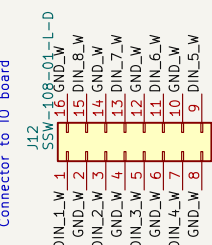


Connector to IO board

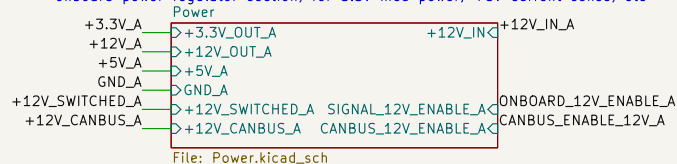
Provides dual fully isolated canbus interfaces (isolated from each-other as well)



Connector to IO board



onboard power regulator section, for 3.3v mcu power, +5v current sense, etc



File: Power.kicad_sch

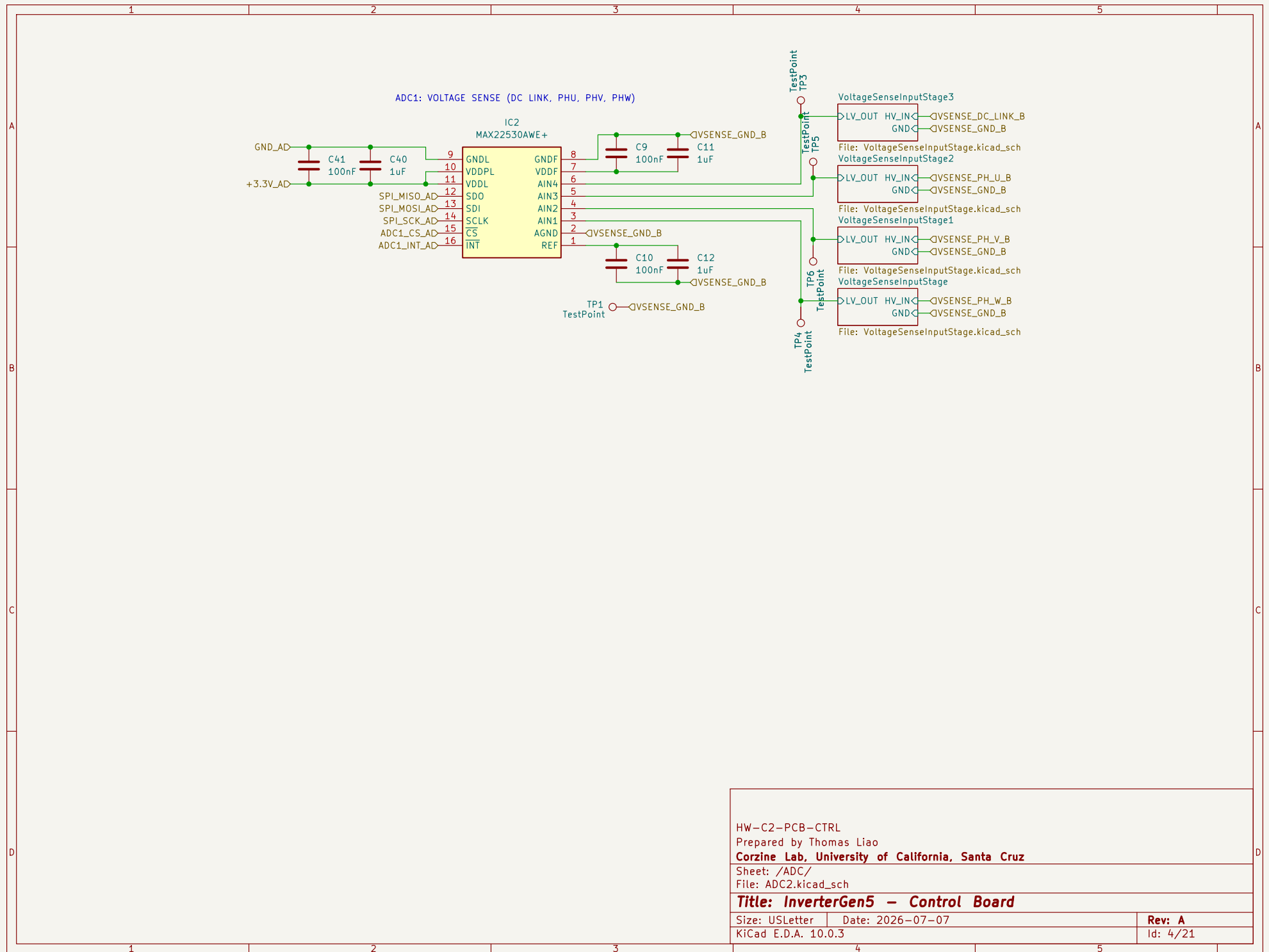
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /
File: ControlBoard.kicad_sch

Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07
KiCad E.D.A. 10.0.3

Rev: A
Id: 1/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

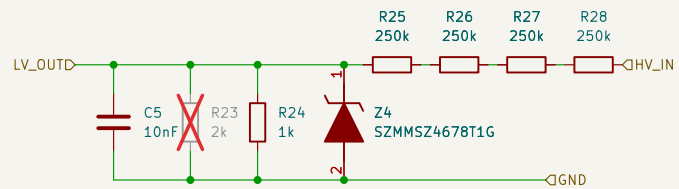
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

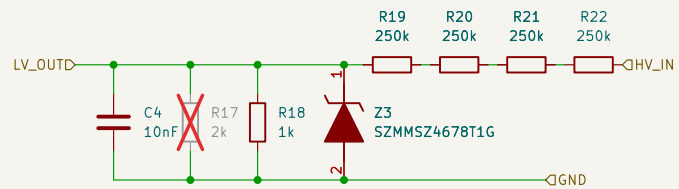
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

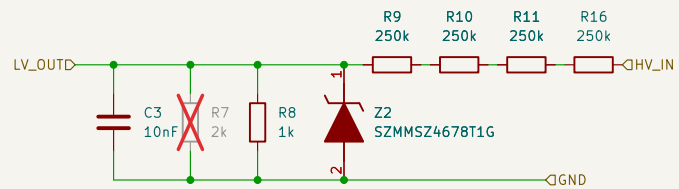
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

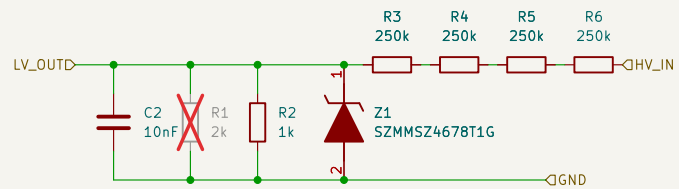
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

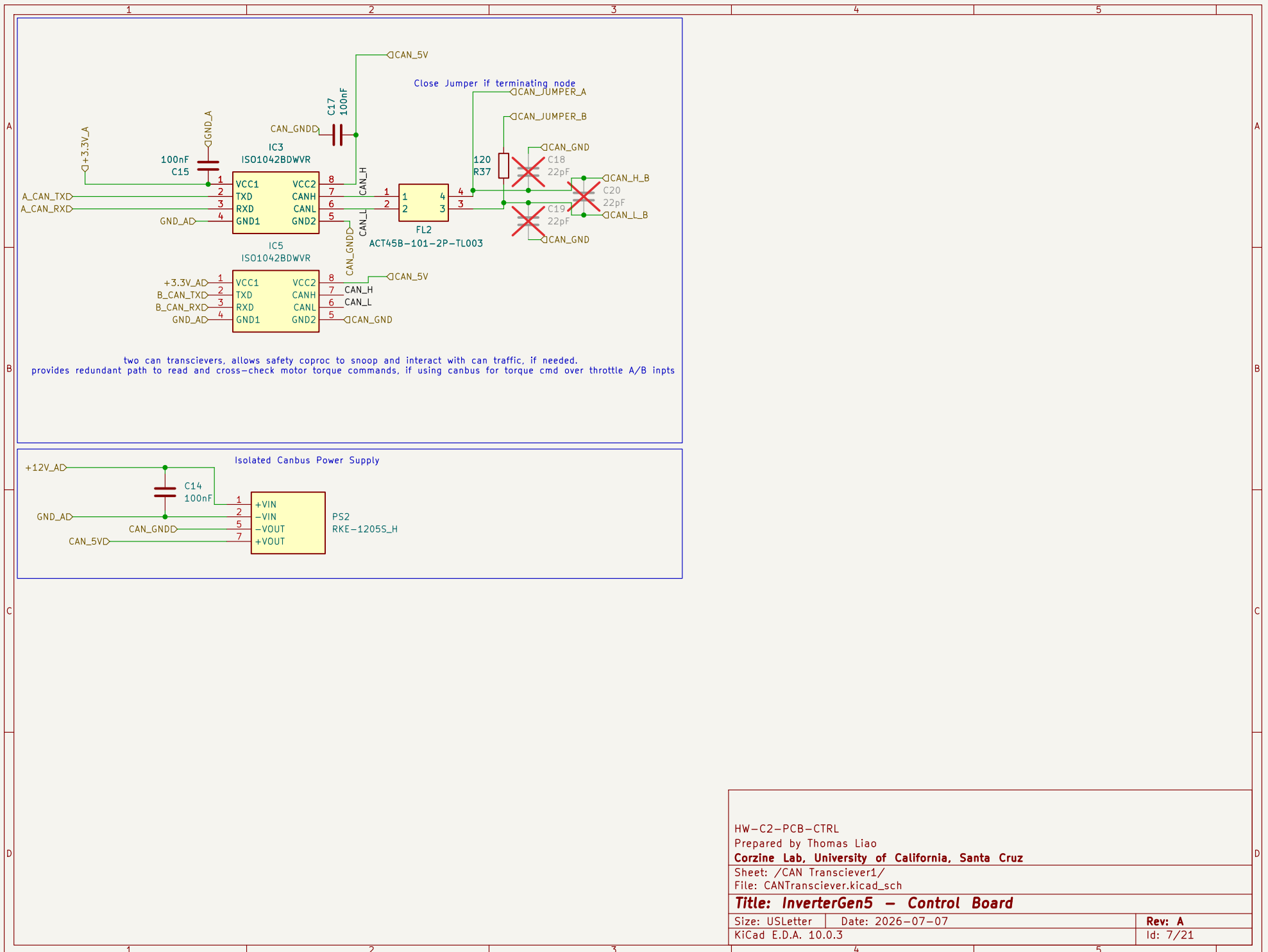
Size: USLetter

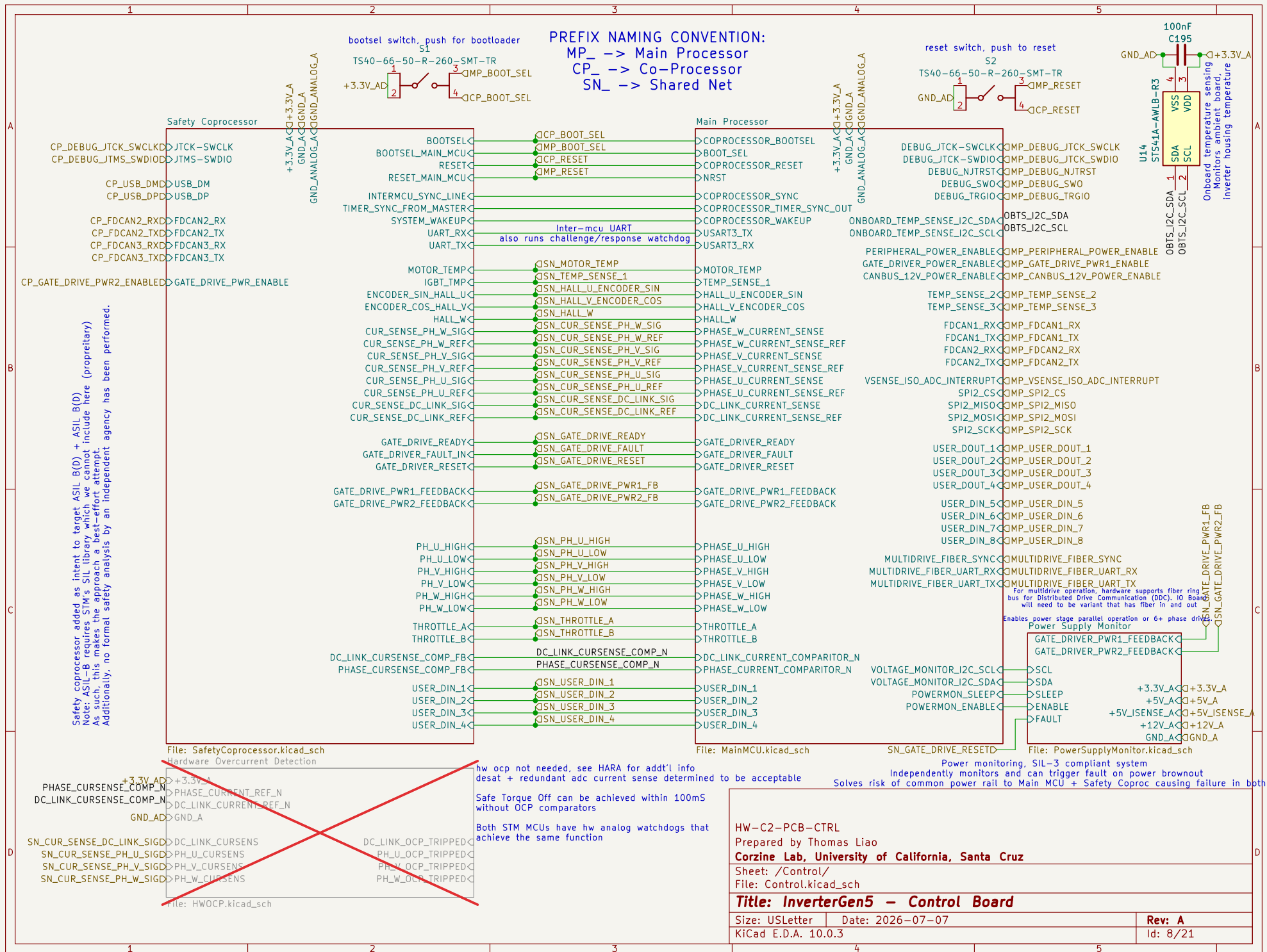
Date: 2026-07-07

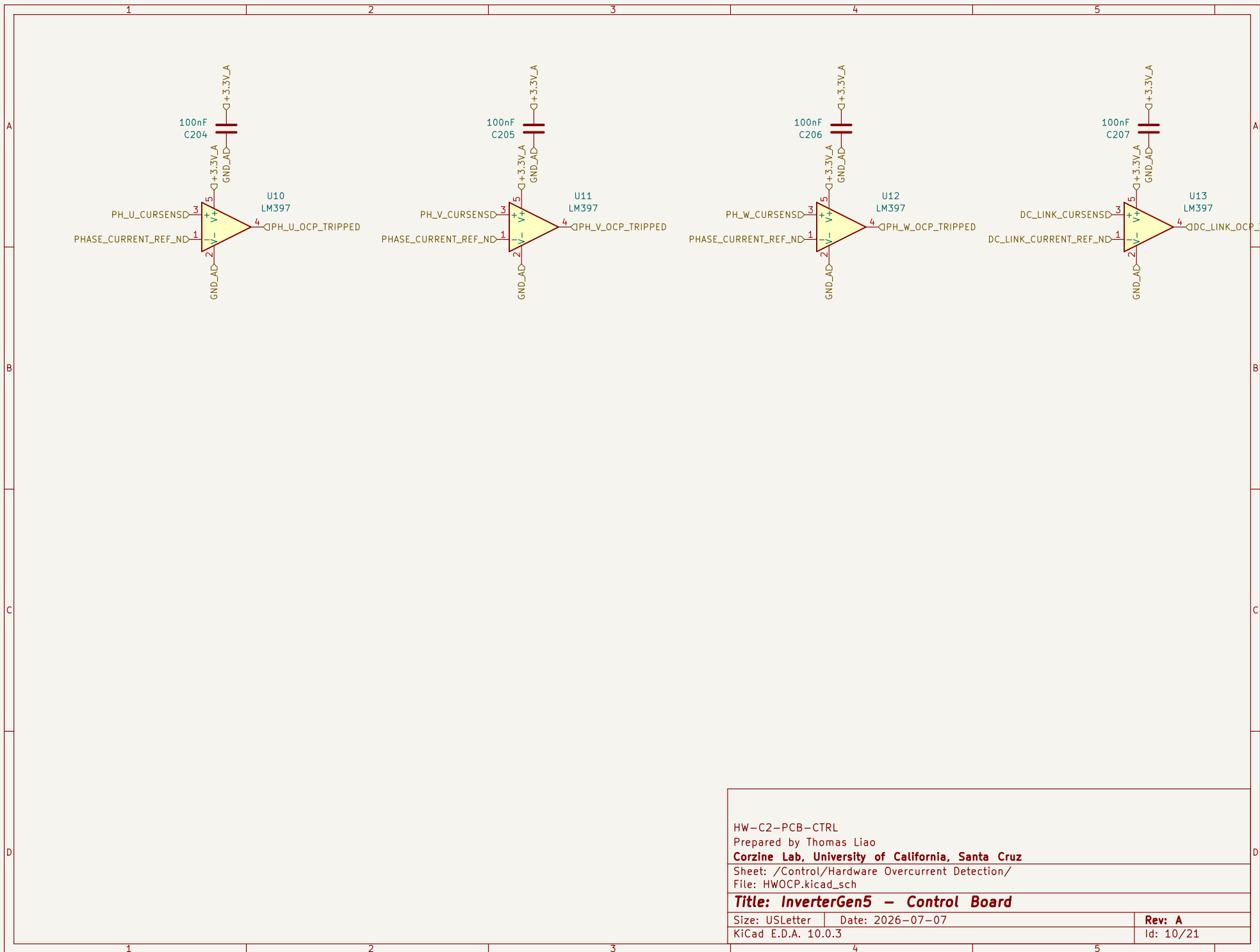
Rev: A

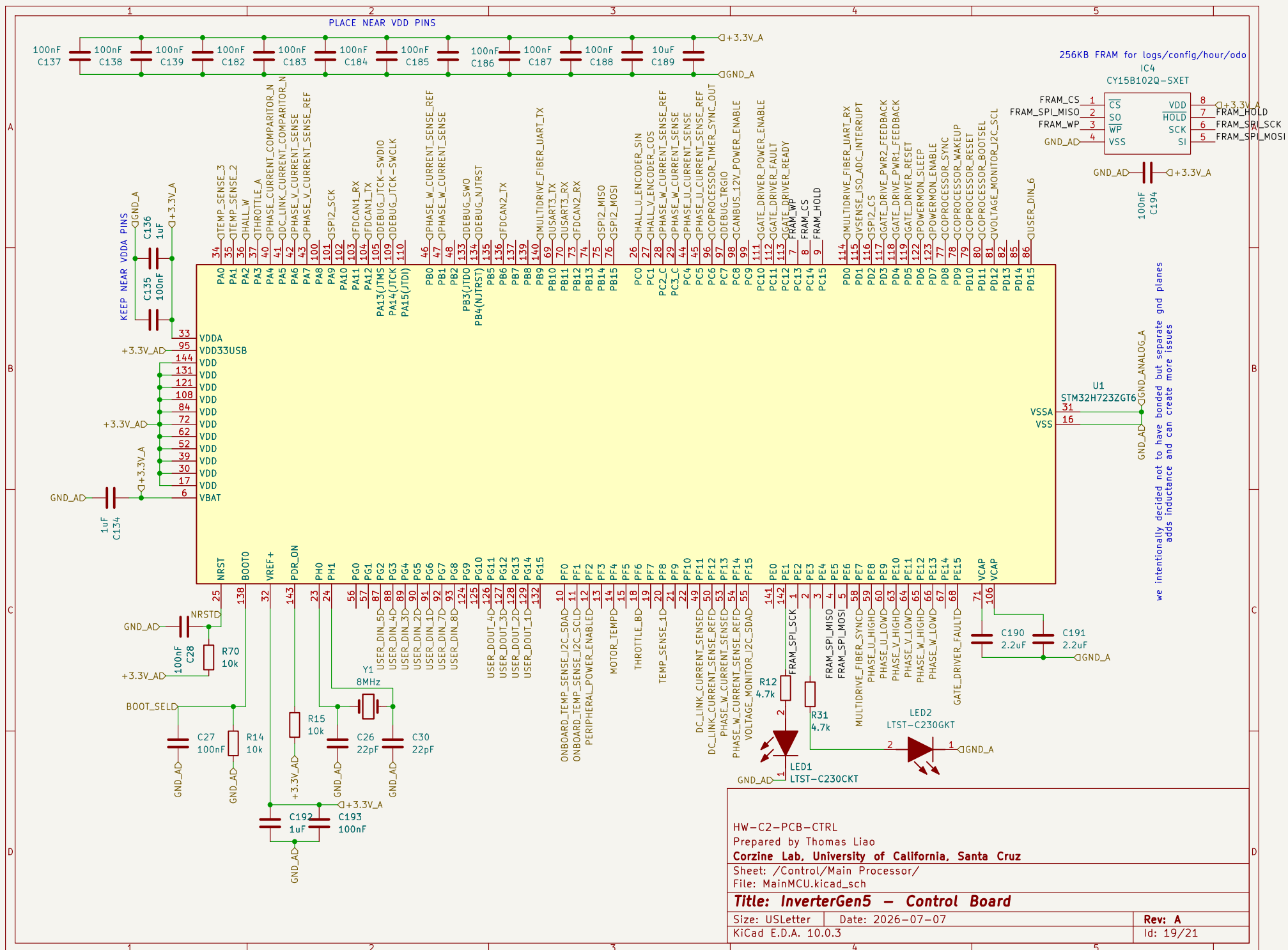
KiCad E.D.A. 10.0.3

Id: 6/21

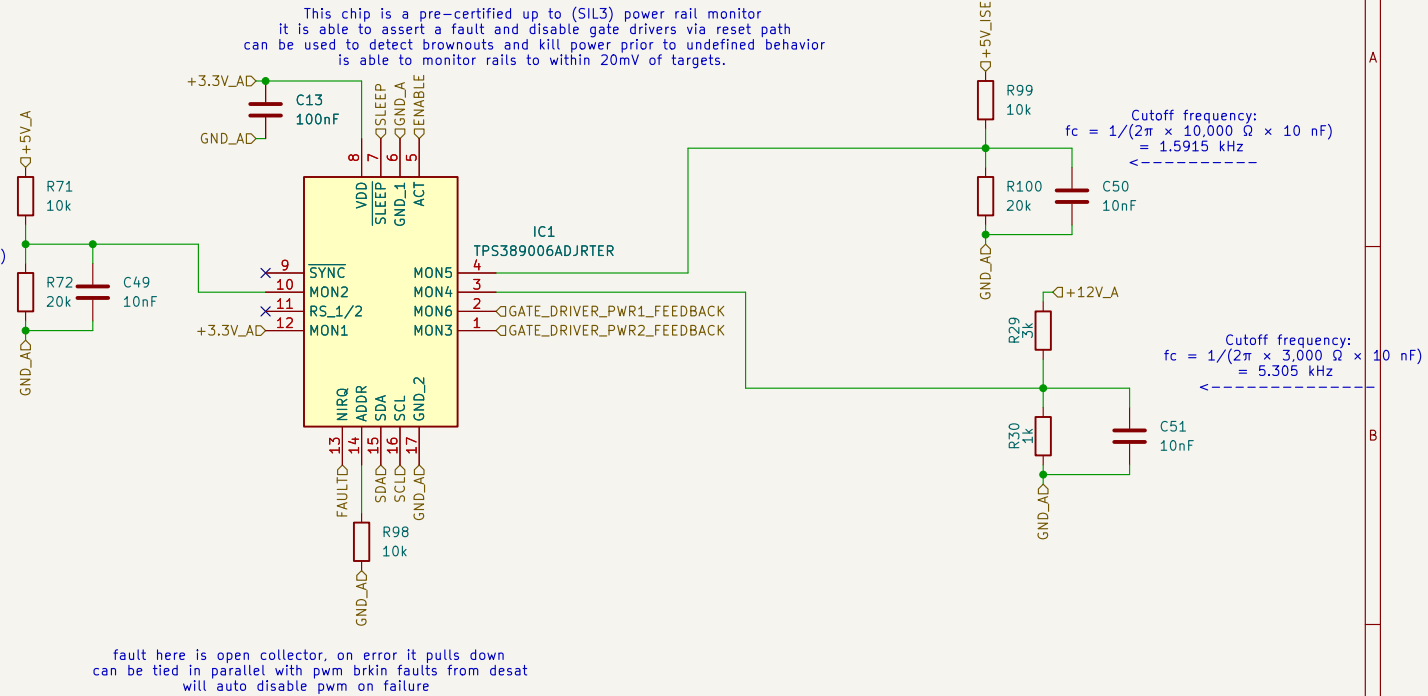








$$\text{Cutoff frequency: } f_c = 1/(2\pi \times 10,000 \, \Omega \times 10 \, \text{nF}) = 1.5915 \, \text{kHz}$$



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/

File: PowerSupplyMonitor.kicad_sch

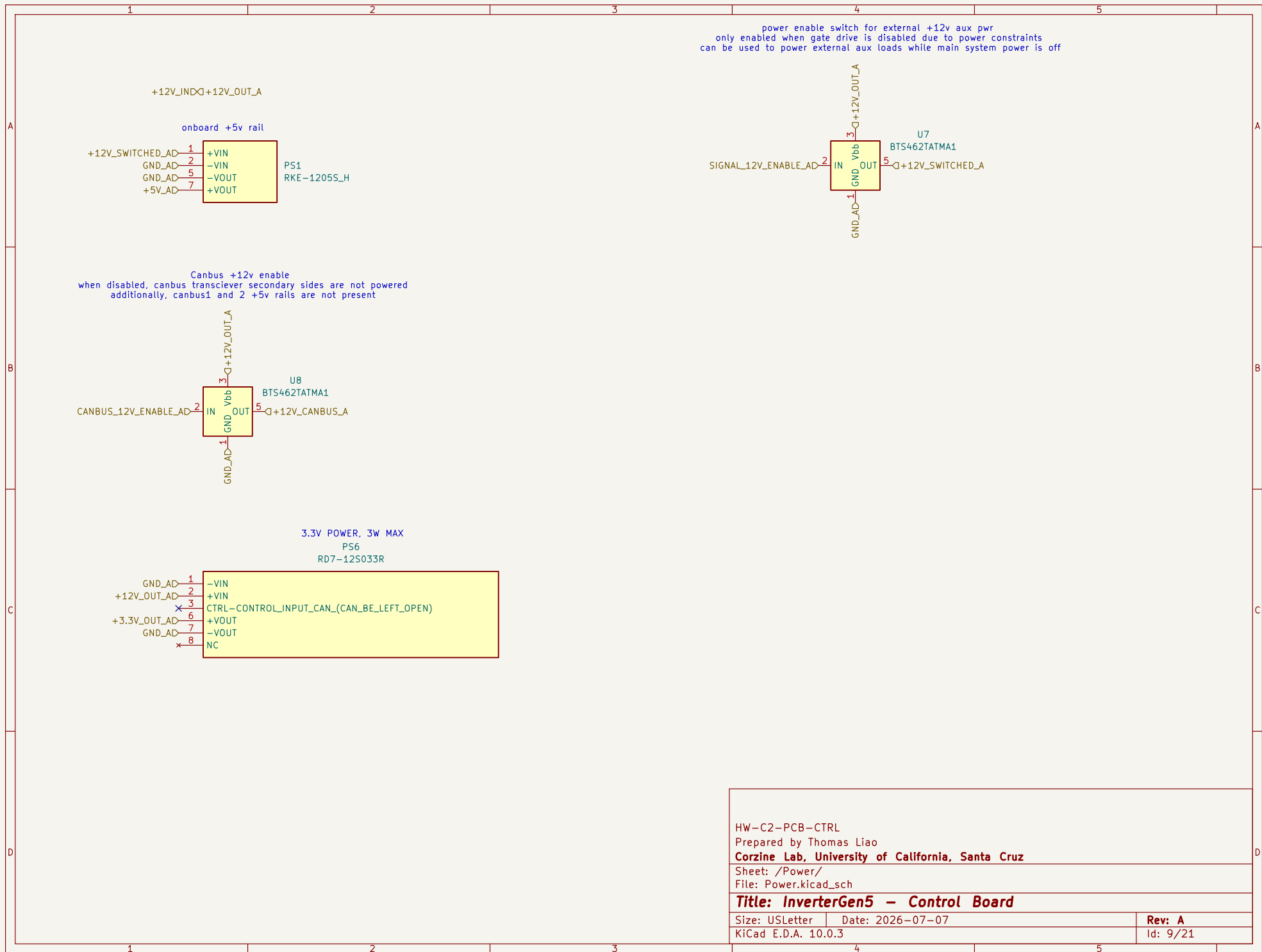
Title: InverterGen5 - Control Board

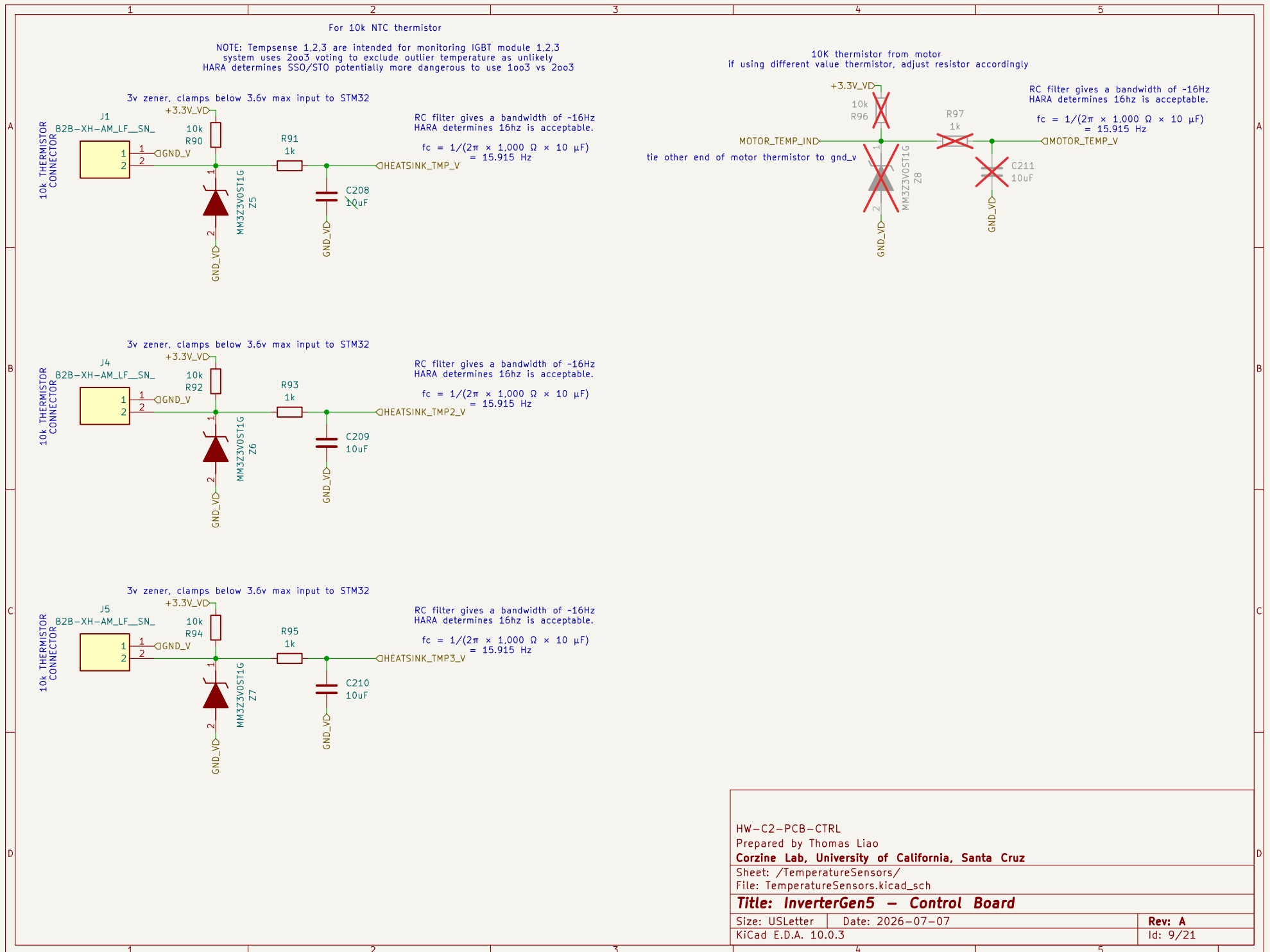
Size: USLetter Date: 2026-07-07

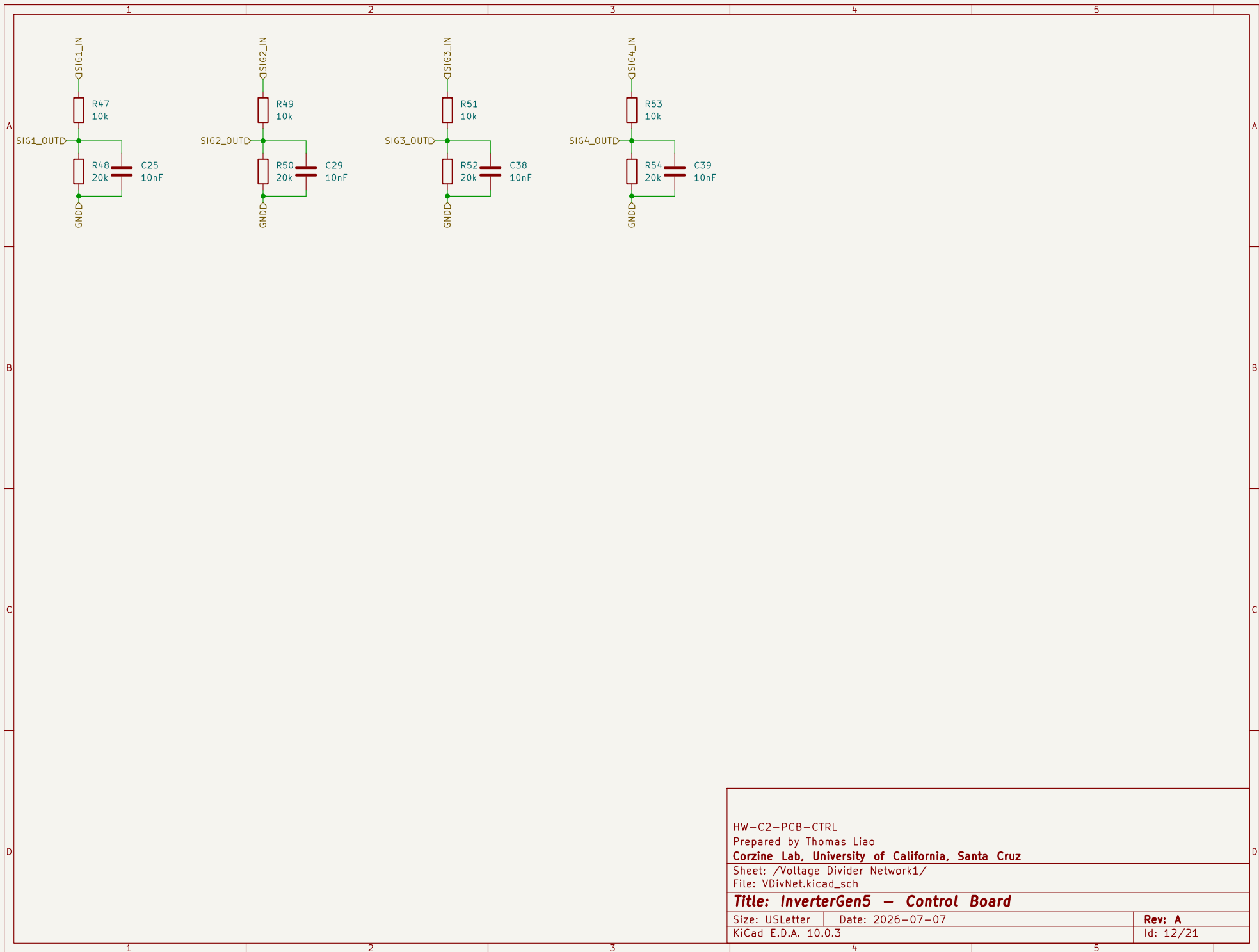
KiCad E.D.A. 10.0.3

Rev: A

Id: 20/21







HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

Title: InverterGen5 - Control Board

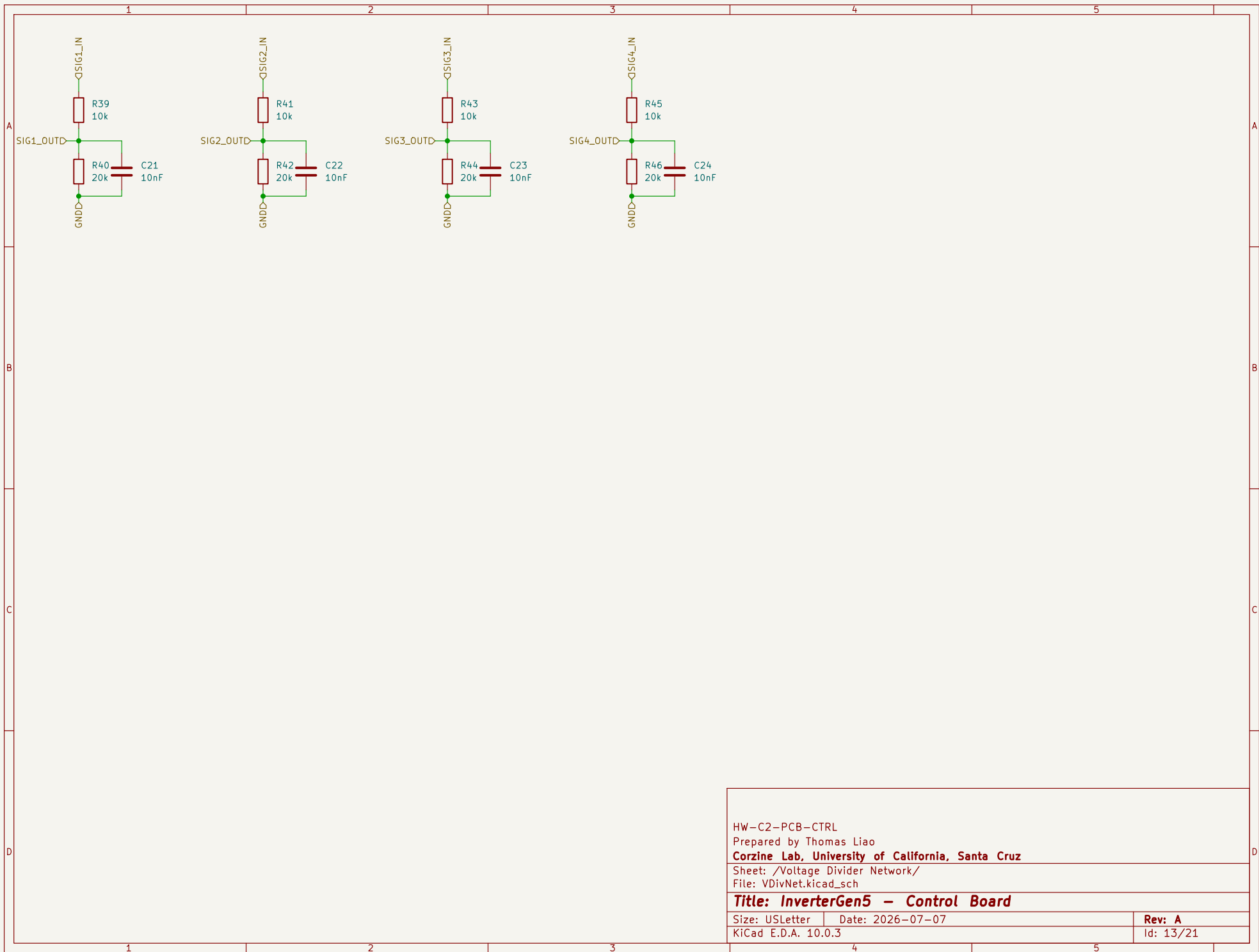
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Date: 2026-07-07

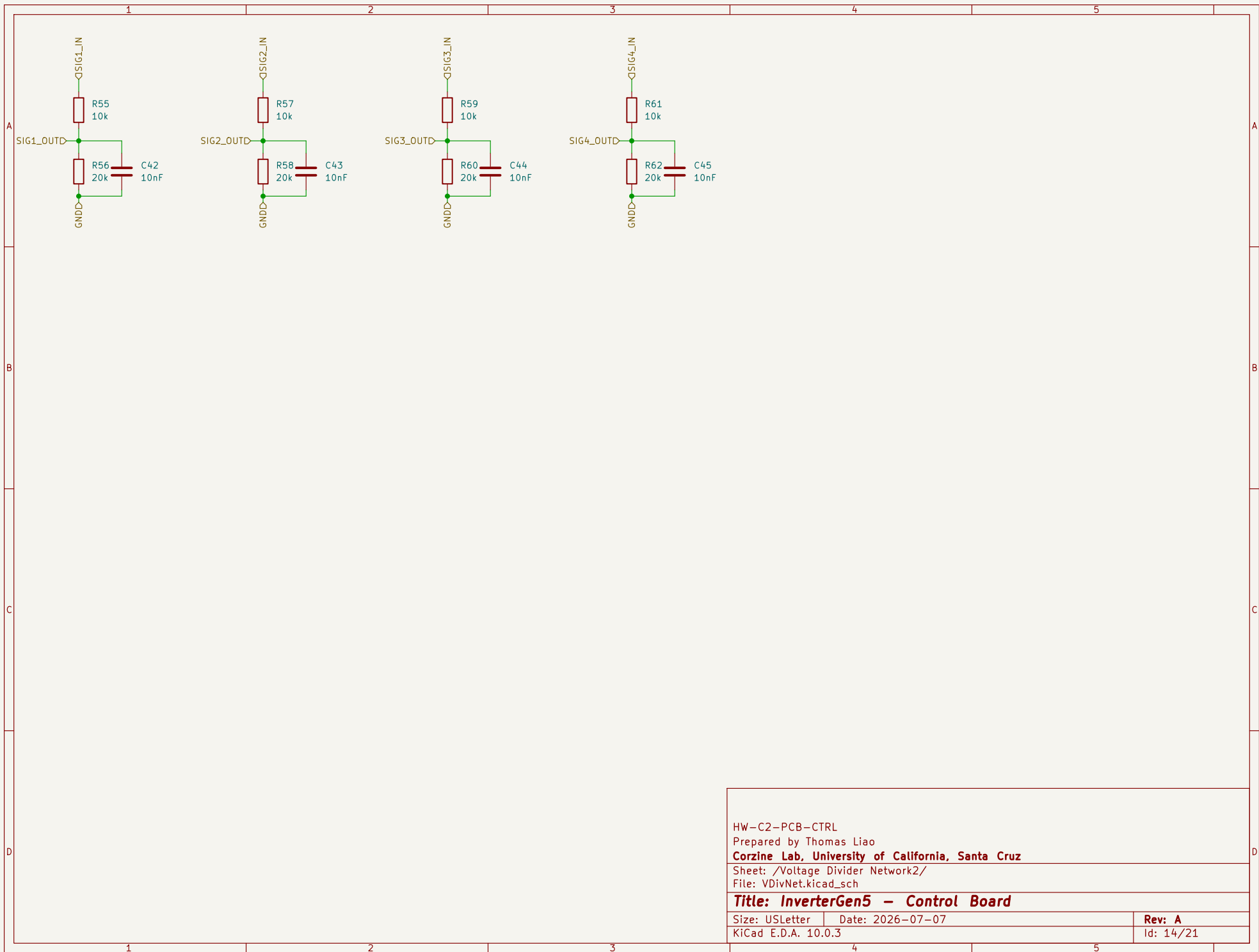
Rev: A

KiCad E.D.A. 10.0.3

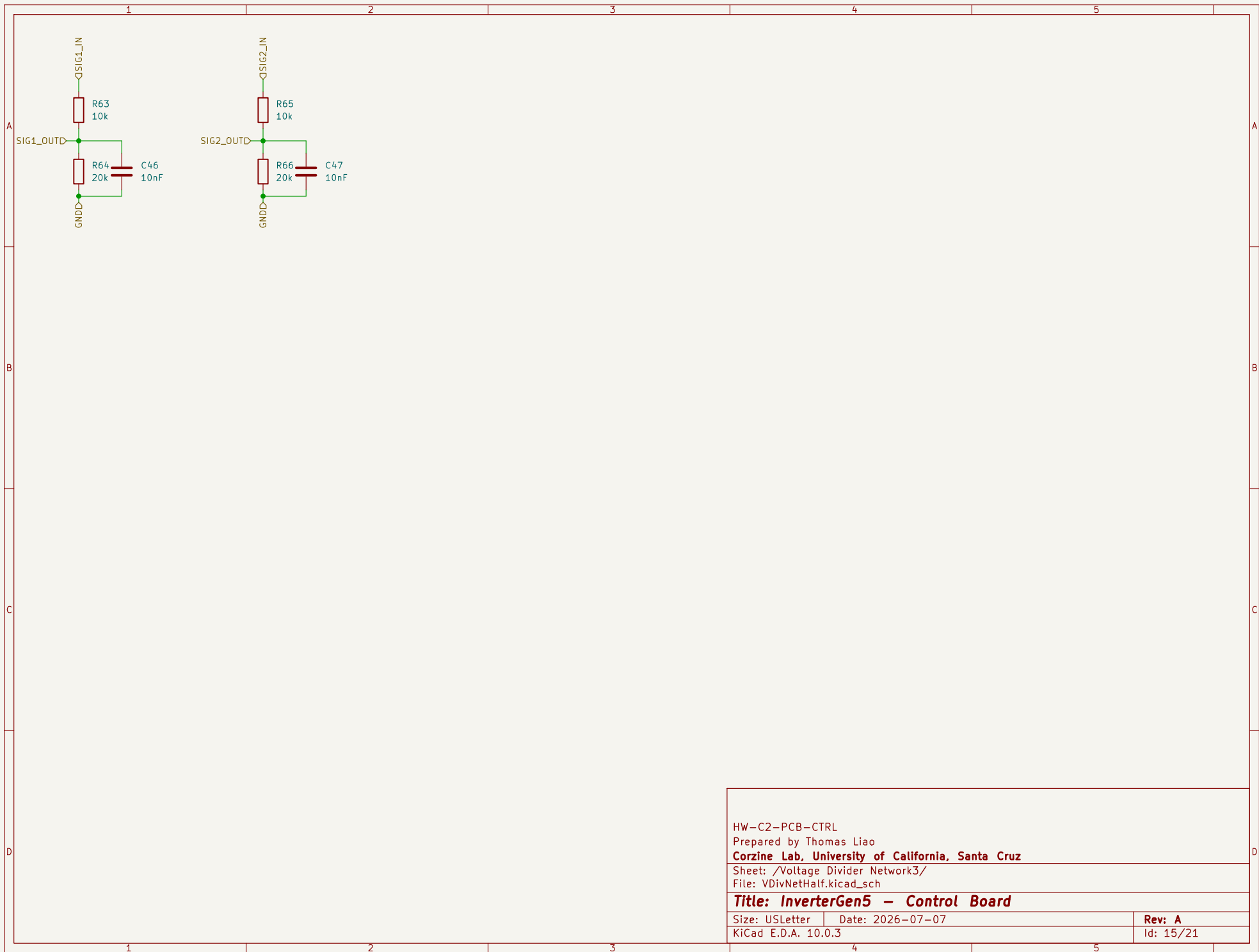
Id: 12/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter | Date: 2026-07-07 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 13/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter | Date: 2026-07-07 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 14/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/

File: VDivNetHalf.kicad_sch

Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 15/21

